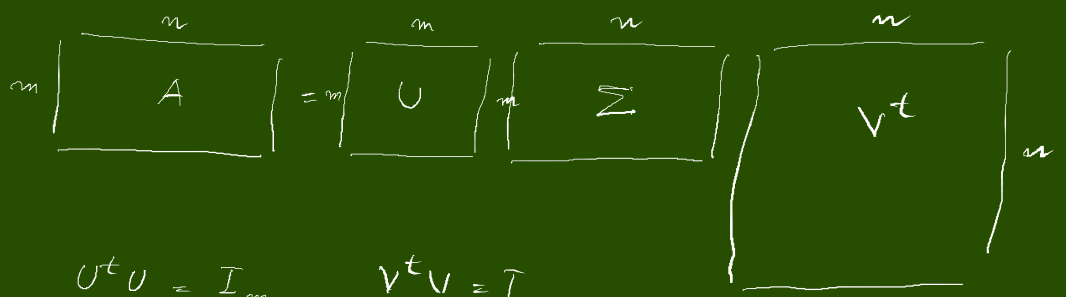
$A \in \mathcal{M}_{m,n}(\mathbb{R})$



$U^t U = I_m \qquad V^t V = I_n$

$$\Sigma = \left(\begin{array}{c|c} \begin{matrix} \sigma_1 & & 0 \\ & \ddots & \\ 0 & & \sigma_n \end{matrix} & 0 \\ \hline 0 & 0 \end{array}\right)$$

$\sigma_1 \geq \dots \geq \sigma_n > 0$ valorile singulare ale matricii A

Algoritm de determinare a SVD:

1) V

$A \in \mathcal{M}_{m,n}(\mathbb{R}) \Longrightarrow A^t A \in \mathcal{M}_n(\mathbb{R})$ matrice simetrică

$(n,m)(m,n)$

(n,n)

$$(A^t A)^t = A^t (A^t)^t = A^t A$$

$\Rightarrow \quad A^t A = V \cdot D \cdot V^t$

$D = \begin{pmatrix} \lambda_1 & & \\ & \ddots & \\ & & \lambda_n \end{pmatrix} \qquad \lambda_1 \geq \dots \geq \lambda_n$

$V = (v_1 | \dots | v_n)$ V ortogonală

$A^t A v_1 = \lambda_1 v_1 \quad \dots \quad A^t A v_n = \lambda_n v_n$!

$\{v_1, \dots, v_n\}$ bază ortonormată în $\mathbb{R}^n$

2) Σ

obs. 1) $\lambda_1 \geq \dots \geq \lambda_n \geq 0$

$\lambda_1 \geq \dots \geq \lambda_r > 0 = \lambda_{r+1} = \dots = \lambda_n$

$A^t A v_i = \lambda_i v_i \Big|_{v_i^t} \quad i = \overline{1,n}$

$v_i^t A^t A v_i = \lambda_i v_i^t v_i$

$\|Av_i\|^2 = \lambda_i \cdot \|v_i\|^2 \Rightarrow \lambda_i \geq 0 \quad \forall i = \overline{1,n}$

2) $r = \operatorname{rang} A$.

$n - r = \dim_{\mathbb{R}} V_{\lambda = 0}$

$V_{\lambda=0} = \{ v \in \mathbb{R}^n \mid A^t A v = 0 . v \}$

$= \{ v \in \mathbb{R}^n \mid A^t A v = 0 \}$

$= \ker(A^t A) = \ker(A)$

$\Rightarrow n - r = \dim_{\mathbb{R}} \ker(A) = n - \operatorname{rang}(A)$

$\Rightarrow r = \operatorname{rang}(A)$

Avec $\sigma_1 = \sqrt{\lambda_1} \geq \dots \geq \sigma_r = \sqrt{\lambda_r} > 0$ val. singulières.

$$\Sigma = \begin{array}{c} \\ r \\ n-r \end{array} \begin{array}{c} \begin{array}{ccc} \sigma_1 & & \\ & \ddots & \\ & & \sigma_r \end{array} \quad \begin{array}{c} n-r \\ \end{array} \\ \hline \begin{array}{ccc} 0 & & \end{array} \quad \begin{array}{c} 0 \end{array} \end{array}$$

3) U

$$\|Av_i\|^2 = \lambda_i \|v_i\|^2 = \lambda_i$$

$$\|Av_i\| = \begin{cases} \sqrt{\lambda_i} = \sigma_i, & i = \overline{1, r} \\ 0, & i = \overline{r+1, n} \end{cases}$$

$$\underline{Av_i = 0}, \quad i = \overline{r+1, n} \Rightarrow v_i \in \ker(A)$$

$n-r$ vectori ortn.,
$\dim \ker(A) = n - r$

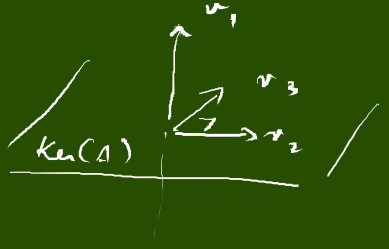
$\Rightarrow$ $\{v_{r+1}, \dots, v_n\}$ bază ortonormată în $\ker(A)$

$\{v_1 \dots v_r\}$ bază ortonormată în $(\ker(A))^{\perp} =$

$\in \mathbb{R}^n$ $\quad \operatorname{Im}(A) \subseteq \mathbb{R}^m$

$= \operatorname{Im}(A^t)$

$\operatorname{Im}(A^t)$



$Av_1, \dots, Av_m$ vectori ortogonali $\Leftrightarrow \langle Av_i, Av_j \rangle = 0$, $i \neq j$

$$\langle Av_i, Av_j \rangle = (Av_i)^t Av_j = v_i^t (A^t A v_j) = v_i^t \lambda_j v_j =$$

$$= \lambda_j \underbrace{\langle v_i, v_j \rangle}_{0} = 0, \quad i \neq j$$

$u_1 = \frac{1}{\|Av_1\|} Av_1, \dots, u_r = \frac{1}{\|Av_r\|} Av_r$ vectori ortonormati

$u_1 = \frac{1}{\sigma_1} Av_1, \dots, u_r = \frac{1}{\sigma_r} Av_r \in \mathrm{Im}(A)$ în $\mathbb{R}^m$

r vectori ortonormati (*)

Completăm (dacă $r < m$) la o bază ortonormată în $\mathbb{R}^m$: $\{u_1, \dots, u_r, u_{r+1}, \dots, u_m\}$

$U = (u_1 | \dots | u_m)$

4) $A = U \Sigma V^t \quad / \cdot V \Leftrightarrow AV = U\Sigma$

$$AV = A(v_1 | \dots | v_n | v_{n+1} | \dots | v_m) =$$

$$= (Av_1 | \dots | Av_n | Av_{n+1} | \dots | Av_m)$$

$$= (\sigma_1 u_1 | \dots | \sigma_n u_n | 0 | \dots | 0) =$$

$$= (u_1 | \dots | u_n | u_{n+1} | \dots | u_m) \cdot \begin{pmatrix} \sigma_1 & & & \\ & \ddots & & 0 \\ & & \sigma_n & \\ \hline & 0 & & 0 \end{pmatrix}$$

$$= U\Sigma \quad \Leftrightarrow \quad A = U\Sigma V^t$$

$\{u_1, \dots u_n\}$ bază ortonormată în $\mathrm{Im}(A)$

$\{u_{n+1}, \dots u_m\}$ bază ortonormată în $\mathrm{Im}(A)^{\perp} = \ker(A^t)$

$$AA^t = U\Sigma \underbrace{V^t V}_{I_m} \Sigma^t U^t = U(\Sigma\Sigma^t)U^t$$

$$\underset{(m,n)(n,m)\ (m,m)}{\Sigma \cdot \Sigma^t} = \begin{pmatrix} \sigma_1 & & & & \\ & \ddots & & 0 & \\ & & \sigma_r & & \\ \hline & 0 & & 0 & \end{pmatrix}_{m-r} \begin{pmatrix} \sigma_1 & & & & \\ & \ddots & & 0 & \\ & & \sigma_r & & \\ \hline & 0 & & 0 & \end{pmatrix}_{n-r}$$

$$= \begin{pmatrix} \sigma_1^2 & & & & \\ & \ddots & & 0 & \\ & & \sigma_r^2 & & \\ & 0 & & \ddots & \\ & & & & 0 \end{pmatrix}_{m}$$

$\underset{(n,n)}{A^tA}$? $\underbrace{AA^t}_{(m,m)}$ ⟶ SVD A^t ⟶ SVD A

$m < n$

Exemple. $A = \begin{pmatrix} 1 & 1 \\ 0 & 0 \\ 2 & 2 \end{pmatrix}$

$$A^tA = \begin{pmatrix} 1 & 0 & 2 \\ 1 & 0 & 2 \end{pmatrix}\begin{pmatrix} 1 & 1 \\ 0 & 0 \\ 2 & 2 \end{pmatrix} = \begin{pmatrix} 5 & 5 \\ 5 & 5 \end{pmatrix} \begin{cases} \lambda_1 = 10 \\ \lambda_2 = 0 \end{cases}$$

$\lambda_1 = 10 \quad V_{\lambda_1} = \{ v \in \mathbb{R}^2 \mid A^t A v = \lambda_1 v \}$

$$\begin{cases} 5x + 5y = 10x \\ 5x + 5y = 10y \end{cases} \Rightarrow x = y$$

$$V_{\lambda_1} = \left\{ \begin{pmatrix} x \\ x \end{pmatrix} \mid x \in \mathbb{R} \right\} = Sp\left\{ \begin{pmatrix} 1 \\ 1 \end{pmatrix} \right\}$$

$\underbrace{\left\{ \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ 1 \end{pmatrix} \right\}}_{v_1}$ base ortonormale in V_{λ_1}

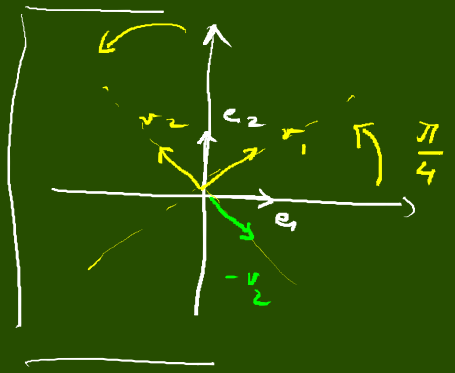
$\lambda_2 = 0 \quad V_{\lambda_2} = \dots$

$$\begin{cases} 5x + 5y = 0 \\ 5x + 5y = 0 \end{cases} \Rightarrow x = -y \quad (y = -x)$$

$$V_{\lambda_2} = \left\{ \begin{pmatrix} -y \\ y \end{pmatrix} \mid y \in \mathbb{R} \right\} = Sp\left\{ \begin{pmatrix} -1 \\ 1 \end{pmatrix} \right\} \quad (Sp\{(-1,1)\})$$

$\underbrace{\left\{ \frac{1}{\sqrt{2}} \begin{pmatrix} -1 \\ 1 \end{pmatrix} \right\}}_{v_2}$ base ortonormale in V_{λ_2}

$v_1 \perp v_2$



V ortogonală $V \in \mathcal{M}_2(\mathbb{R})$

V matrice de rotație $\Rightarrow \det V = 1$

$(V^t V = I_2 \Rightarrow (\det V)^2 = 1)$

$$\Rightarrow \quad V = (v_1 / v_2) = \begin{pmatrix} 1/\sqrt{2} & -1/\sqrt{2} \\ 1/\sqrt{2} & 1/\sqrt{2} \end{pmatrix}$$

$\lambda_1 = 10 \qquad \lambda_2 = 0$

$\Downarrow$

$\sigma_1 = \sqrt{10}$

$\Downarrow$

$$\Sigma' = \begin{pmatrix} \sqrt{10} & 0 \\ 0 & 0 \\ 0 & 0 \end{pmatrix}$$

$A, \Sigma \in \mathcal{M}_{3,2}(\mathbb{R})$

$U = ?$

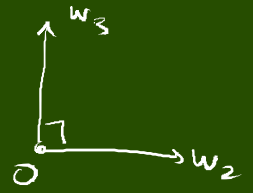
$$u_1 = \frac{1}{\sigma_1} A v_1 = \frac{1}{\sqrt{10}} \begin{pmatrix} 1 & 1 \\ 0 & 0 \\ 2 & 2 \end{pmatrix} \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ 1 \end{pmatrix} = \frac{1}{2\sqrt{5}} \begin{pmatrix} 2 \\ 0 \\ 4 \end{pmatrix}$$

$$u_1 = \frac{1}{\sqrt{5}} \begin{pmatrix} 1 \\ 0 \\ 2 \end{pmatrix} \in \mathbb{R}^3$$

Completo u_1 a una base ortonormal de $\mathbb{R}^3$

$$\det \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 2 & 0 & 1 \end{pmatrix} \neq 0$$

w_1 w_2 w_3



GS:

$$z_1 = w_1 = \begin{pmatrix} 1 \\ 0 \\ 2 \end{pmatrix} \Rightarrow u_1 = \frac{1}{\sqrt{5}} \begin{pmatrix} 1 \\ 0 \\ 2 \end{pmatrix} = \frac{1}{\|z_1\|} z_1$$

$$z_2 = w_2 - \text{proy}_{z_1} w_2 = \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix} - \frac{0}{5} \begin{pmatrix} 1 \\ 0 \\ 2 \end{pmatrix} = \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix} \Rightarrow$$

$$\Rightarrow u_2 = \frac{1}{\|z_2\|} z_2 = \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix}$$

$$z_3 = w_3 - \text{proy}_{z_1} w_3 - \text{proy}_{z_2} w_3 =$$

$$= \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix} - \frac{2}{5}\begin{pmatrix} 1 \\ 0 \\ 2 \end{pmatrix} - \frac{0}{1}\begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix}$$

$$= \begin{pmatrix} -2/5 \\ 0 \\ 1/5 \end{pmatrix} \Rightarrow u_3 = \frac{1}{\sqrt{5}}\begin{pmatrix} -2 \\ 0 \\ 1 \end{pmatrix} = \frac{1}{\|z_3\|} z_3$$

$$U = (u_1 \mid u_2 \mid u_3) = \begin{pmatrix} 1/\sqrt{5} & 0 & -2/\sqrt{5} \\ 0 & 1 & 0 \\ 2/\sqrt{5} & 0 & 1/\sqrt{5} \end{pmatrix}$$

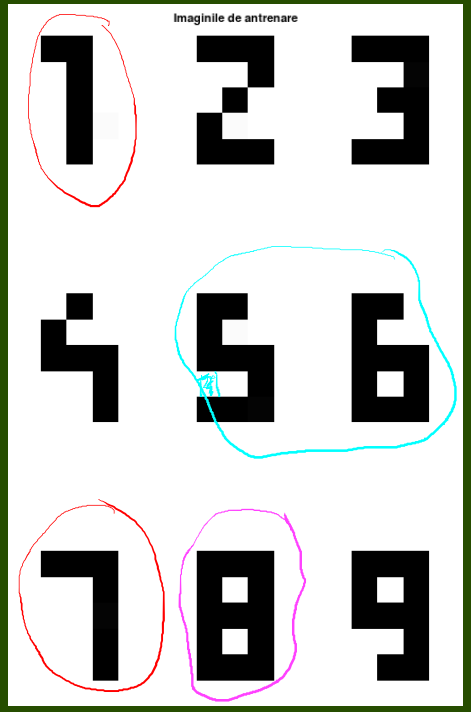
$$\begin{pmatrix} 1 & 1 \\ 0 & 0 \\ 2 & 2 \end{pmatrix} = U \cdot \begin{pmatrix} \sqrt{10} & 0 \\ 0 & 0 \\ 0 & 0 \end{pmatrix} \cdot \underbrace{\begin{pmatrix} 1/\sqrt{2} & 1/\sqrt{2} \\ -1/\sqrt{2} & 1/\sqrt{2} \end{pmatrix}}_{V^t}$$

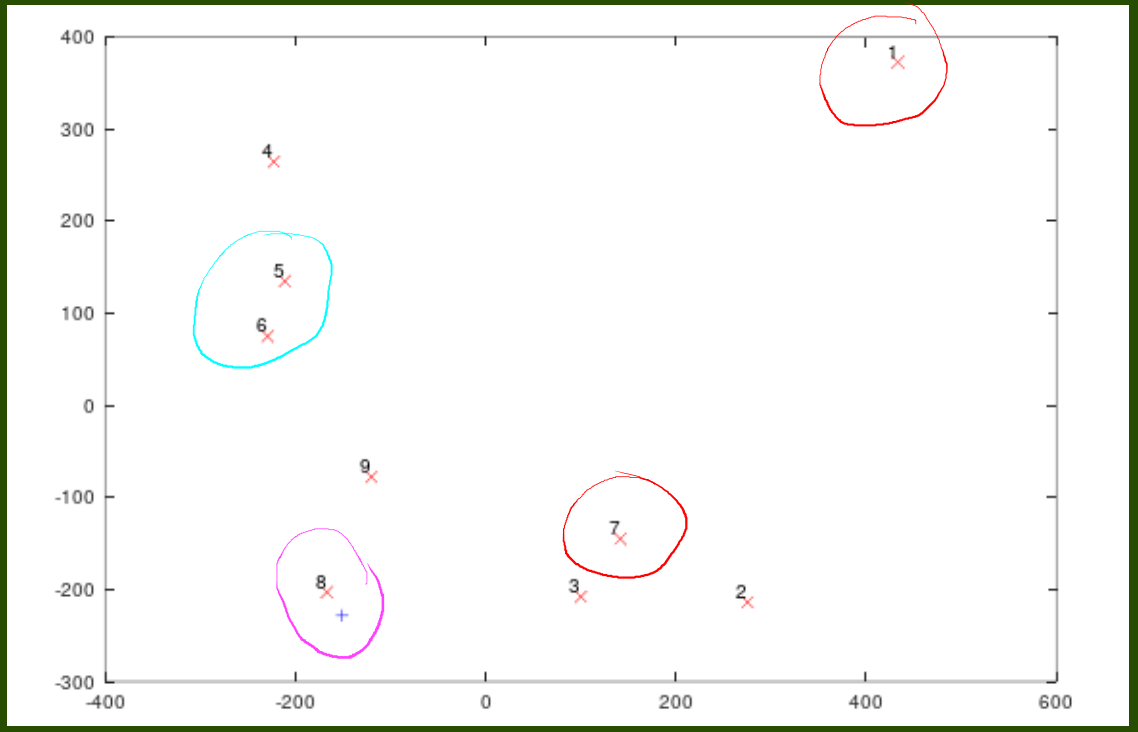
$$A = U \cdot \Sigma \cdot V^t$$

$$A^t A = V \cdot \begin{pmatrix} \sqrt{10} & 0 \\ 0 & 0 \end{pmatrix} \cdot V^t$$

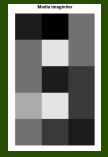
$$A A^t = U \cdot \begin{pmatrix} \sqrt{10} & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix} \cdot U^t$$

$$A = \sigma_1 u_1 v_1^t = \frac{1}{\sqrt{10}} \begin{pmatrix} 1/\sqrt{5} \\ 0 \\ 2/\sqrt{5} \end{pmatrix} \begin{pmatrix} 1/\sqrt{2} & 1/\sqrt{2} \end{pmatrix}$$


Imaginile de antrenare


400
300
200
100
0
-100
-200
-300
-400
-200
0
200
400
600
1
2
3
4
5
6
7
8
9

media


Media imaginilor


Imaginea test

dreaptă

$f(x) = ax + b$ $\qquad f(x,y) = ax + by + c \in \mathbb{R}$

$f : \mathbb{R}^2 \longrightarrow \mathbb{R}$

x^2, y^2, xy

$q(v) = \langle Av, v \rangle = (Av)^t v = v^t A^t v = v^t A v$

$A = \begin{pmatrix} 5 & 2 \\ 2 & 3 \end{pmatrix} \longleftrightarrow q\begin{pmatrix} x \\ y \end{pmatrix} = 5x^2 + 3y^2 + 4xy$ $\qquad 2+2$

$(x\ y)\begin{pmatrix} 5 & 2 \\ 2 & 3 \end{pmatrix}\begin{pmatrix} x \\ y \end{pmatrix} =$

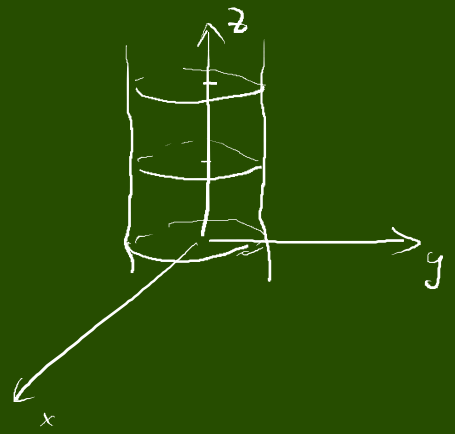
$A = \begin{pmatrix} 5 & 2 \\ 2 & 3 \end{pmatrix}$ $\qquad P_A(X) = X^2 - 8X + 11$

$\Delta = 64 - 44 = 20.$

$\lambda_{1,2} = \dfrac{8 \pm 2\sqrt{5}}{2} = 4 \pm \sqrt{5}$

$\max_{x^2+y^2=1} \underbrace{\left(5x^2 + 3y^2 + 4xy\right)}_{q\begin{pmatrix} x \\ y \end{pmatrix}} = 4 + \sqrt{5}$

• $\min\left(\frac{}{}\right) = 4 - \sqrt{5}$



1 ec. plan / suprafață

2 ec. dreaptă / curbă

$x^2 + y^2 = 1$